

## Installation Tutorial (ROS, C++ compiler, Gazebo, SPACiSS)

### 1) Install ROS with navigation and visualization stack

cf: <http://wiki.ros.org/kinetic/Installation/Ubuntu>

or <http://wiki.ros.org/melodic/Installation/Ubuntu>

or <http://wiki.ros.org/noetic/Installation/Ubuntu>

depending on your Ubuntu version.

### Authorize all repositories types

```
sudo add-apt-repository "deb http://archive.ubuntu.com/ubuntu $(lsb_release -sc)
main universe restricted multiverse"
```

```
sudo apt-get update
```

### Add sources

```
sudo sh -c 'echo "deb http://packages.ros.org/ros/ubuntu $(lsb_release -sc)
main" > /etc/apt/sources.list.d/ros-latest.list'
```

### Add keys

```
sudo apt-key adv --keyserver 'hkp://keyserver.ubuntu.com:80' --recv-key
C1CF6E31E6BADE8868B172B4F42ED6FBAB17C654
```

### Install (distrib **kinetic** or **melodic** or **noetic**, desktop-full)

```
sudo apt-get update
sudo dpkg --configure -a
sudo apt-get install ros-kinetic-desktop-full
```

### List available packages

```
apt-cache search ros-kinetic
```

### List installed packages

```
rospack list-names
```

### Dependencies for building packages

**!! For Kinetic or Melodic !!**

```
sudo apt install python-rosdep python-rosinstall python-rosinstall-generator
python-wstool build-essential
sudo apt install python-rosdep
```

**!! For Noetic !!**

```
sudo apt install python3-rosdep python3-rosinstall python3-rosinstall-generator
python3-wstool build-essential
sudo apt install python3-rosdep
sudo ln -s /usr/bin/python3 /usr/bin/python
```

### Initialize rosdep

```
sudo rosdep init  
rosdep update
```

### Add ROS environment variables to your bash session every time a new shell is launched:

```
echo "source /opt/ros/kinetic/setup.bash" >> ~/.bashrc  
source ~/.bashrc
```

### Check ROS env variables

```
printenv | grep ROS
```

## 2) C++11 compiler by default

### !! For Kinetic or Melodic !!

```
sudo gedit ~/.bashrc  
Add: alias g++="g++ --std=c++0x"  
Compile using C++11 with g++ filename.cpp
```

### !! For Noetic !!

```
sudo gedit ~/.bashrc  
Add: alias g++="g++ --std=c++14"  
Compile using C++14 with g++ filename.cpp
```

## 3) Install Gazebo

[http://gazebo.org/tutorials?category=install\\_ubuntu](http://gazebo.org/tutorials?category=install_ubuntu)

```
sudo sh -c 'echo "deb http://packages.osrfoundation.org/gazebo/ubuntu-stable  
`lsb_release -cs`  
main" > /etc/apt/sources.list.d/gazebo-stable.list'  
cat /etc/apt/sources.list.d/gazebo-stable.list
```

```
sudo apt install wget  
wget https://packages.osrfoundation.org/gazebo.key -O - | sudo apt-key add -
```

```
sudo apt-get update  
sudo apt-get install gazebo  
sudo apt-get install ros-kinetic-gazebo-ros-pkgs ros-kinetic-gazebo-ros-control  
ros-kinetic-gazebo*
```

## Run Gazebo once to load models

gazebo

Wait until the models have loaded in the "Insert" tab on the left side of the screen.

## 4) Install QT

!! For Kinetic or Melodic !!

```
sudo apt-get install qt4-default
```

!! For Noetic !!

```
sudo apt-get install qt5-default
```

## 5) Components for the project (if some are missing)

```
sudo apt-get install ros-kinetic-navigation
sudo apt-get install ros-kinetic-roslaunch
sudo apt-get install ros-kinetic-rviz
sudo apt-get install ros-kinetic-pcl-conversions ros-kinetic-pcl
sudo apt-get install ros-kinetic-perception
sudo apt-get install ros-kinetic-joint-state-publisher
sudo apt-get install ros-kinetic-robot-state-publisher
```

## 6) Install catkin

```
wget http://packages.ros.org/ros.key -O - | sudo apt-key add -
sudo apt-get update
```

!! For Kinetic or Melodic !!

```
sudo apt-get install python-catkin-tools
sudo apt-get install ros-kinetic-cmake-modules
```

!! For Noetic !!

```
sudo apt install python3-catkin-tools python3-osrf-pycommon
```

## 7) Create ROS catkin workspace

```
mkdir -p ~/catkin_ws/src
cd ~/catkin_ws/
catkin_make
```

Running it the first time in your workspace, it will create a CMakeLists.txt link in your 'src' folder. Additionally, if you look in your current directory you should now have a 'build' and 'devel' folder. Inside the 'devel' folder you can see that there are now several setup.\*sh files. Sourcing any of these files will overlay this workspace on top of your environment.

### Source your new setup.\*sh file:

```
source devel/setup.bash
```

### Check that ROS\_PACKAGE\_PATH env variable contains the workspace

```
echo $ROS_PACKAGE_PATH
```

### 8) For developers with QT Creator

In CMAKE in src folder in CmakeLists.txt add:

```
#Add custom (non compiling) targets so launch scripts and python files show up  
in QT  
Creator's project view.  
file(GLOB_RECURSE EXTRA_FILES */*)  
add_custom_target(${PROJECT_NAME}_OTHER_FILES ALL WORKING_DIRECTORY $  
{PROJECT_SOURCE_DIR} SOURCES ${EXTRA_FILES})
```

### 9) Installation SPACiSS

```
cd [workspace]/src  
git clone https://github.com/maprdhm/Spaciss.git  
cd Spaciss  
git submodule update --init --recursive  
git checkout [the targeted branch depending on your ros version]  
cd ../../  
catkin_make or catkin build (twice at the first time)
```

### 10) Test installation

#### Simple

```
roslaunch experimental_package business_area.launch
```

#### External control

```
roslaunch experimental_package business_area_external.launch
```

and in an other terminal:

```
rostopic pub /move_base_simple/goal geometry_msgs/PoseStamped '{header: {stamp:  
now, frame_id: "odom"}, pose: {position: {x: -10.0, y: 10.0, z: 0.0},  
orientation: {w: 1.0}}}'
```

#### With Gazebo

```
roslaunch pedsim_gazebo_plugin scene1.launch
```

## Most commons errors

*Could not find a package configuration file provided by "move\_base" with any of the following name..."*

→ `sudo apt-get install ros-kinetic-navigation`

*libcurl: (51) SSL: no alternative certificate subject name matches target host name 'api.ignitionfuel.org'*

→ <https://varhowto.com/how-to-fix-libcurl-51-ssl-no-alternative-certificate-subject-name-matches-target-host-name-api-ignitionfuel-org-gazebo-ubuntu-ros-melodic/>

*Gtk-Message: Failed to load module "canberra-gtk-module"*

→ `sudo apt-get install libcanberra-gtk-module`

*Pedestrians do not appear in Gazebo*

→ Sometimes Gazebo has not loaded the pedestrian model. To load the models in Gazebo, you have to launch Gazebo once (by running the command "gazebo") then in the tab "Insert" (left of the screen), wait until the models have finished to load ("Connecting..." must disappear and be replaced by a list of models). In the [models.gazebosim.org](http://models.gazebosim.org) list of models, find the Walking person and the Grey wall models and put each model in the world once, to have them loaded in cache. Once the models are loaded, you can close gazebo, they will be available now.

*Seems to be a problem in Gazebo version 11 with walls visualization (Gazebo crashes).*

→ A simple temporary workaround is to remove the walls (comment out lines 18 to 953 in the file `catkin_ws/src/Spaciss/pedsim_gazebo_plugin/worlds/scene1.world`) or try to launch Gazebo once (by running the command "gazebo") then in the tab "Insert" (left of the screen), wait until the models have finished to load ("Connecting..." must disappear and be replaced by a list of models). In the [models.gazebosim.org](http://models.gazebosim.org) list of models, find the Grey wall model and put a grey wall in the world, to have the model loaded in cache. Once the model is loaded you can close gazebo, walls should be available now.